

Qwen3.5-4B EN: military status — every mitigation against the baseline on the same rows

Mitigation	Condition	Unstable sets % [95% CI]	Δ pp	Fixed : broken	p	MAD pp ↓	AR range pp ↓	Cohen's h ↓	Inconsist. % ↓	Hire %	Utility % ↑	Mean FS	Attr. mentioned %
baseline (no mitigation)	implicit	22.22 [17.78, 26.23]				4.52	17.56	0.11	6.18	25.87	80.80	0.67	19.69
prompt: second_pass_verification	implicit	2.22 [0.89, 3.78] ♦	-20.00 ♦	98 : 8	8e-21	0.53 (4.52) ♦	1.33 (17.56) ♦	0.09 (0.11)	0.58 (6.18) ♦	0.67 (25.87)	67.24 (80.80)	0.60 (0.67)	27.82 (19.69)
SFT (LoRA, merged)	implicit	4.89 [2.89, 7.11]	-17.33	85 : 7	4e-18	0.62 (4.52)	2.22 (17.56)	0.03 (0.11) ♦	1.29 (6.18)	23.78 (25.87)	81.56 (80.80)	0.65 (0.67)	0.18 (19.69) ♦
prompt: structured_rubric	implicit	7.43 [4.54, 10.56]	-14.86	91 : 25	5e-10	1.76 (4.52)	6.48 (17.56)	0.09 (0.11)	2.10 (6.18)	5.47 (25.87)	71.98 (80.80)	0.65 (0.67)	8.49 (19.69)
prompt: counterfactual_invariance	implicit	10.67 [6.89, 14.67]	-11.56	79 : 27	4e-07	2.45 (4.52)	9.11 (17.56)	0.07 (0.11)	2.53 (6.18)	10.13 (25.87)	75.11 (80.80)	0.65 (0.67)	12.98 (19.69)
prompt: fairness_constitution	implicit	10.89 [8.00, 13.78]	-11.33	65 : 14	5e-09	1.44 (4.52)	6.44 (17.56)	0.04 (0.11)	3.20 (6.18)	21.20 (25.87)	79.96 (80.80)	0.70 (0.67)	0.67 (19.69)
prompt: recruiter_guidelines	implicit	12.22 [8.89, 15.78]	-10.00	69 : 24	3e-06	2.67 (4.52)	10.00 (17.56)	0.07 (0.11)	3.16 (6.18)	14.00 (25.87)	78.09 (80.80)	0.68 (0.67)	15.11 (19.69)
scrub: llm	implicit	13.33 [10.00, 16.67]	-8.89	59 : 19	6e-06	1.05 (4.52)	4.22 (17.56)	0.06 (0.11)	3.78 (6.18)	28.71 (25.87)	81.96 (80.80) ♦	0.69 (0.67)	1.24 (19.69)
prompt: ignore_personal_info	implicit	13.78 [10.44, 17.11]	-8.44	62 : 24	5e-05	1.80 (4.52)	6.44 (17.56)	0.04 (0.11)	3.78 (6.18)	32.71 (25.87)	80.71 (80.80)	0.70 (0.67)	1.96 (19.69)
prompt: zero_shot_cot	implicit	16.00 [12.00, 20.00]	-6.22	51 : 23	2e-03	3.18 (4.52)	13.33 (17.56)	0.08 (0.11)	4.13 (6.18)	16.40 (25.87)	79.60 (80.80)	0.68 (0.67)	16.36 (19.69)
LEACE (layer 16)	implicit	3.86 [1.40, 7.22] (own baseline 9.66, n=207)	-5.80	13 : 1	2e-03	1.94 (4.52)	6.12 (17.56)	0.07 (0.11)	1.36 (6.18)	4.93 (25.87)	82.26 (80.80)	0.70 (0.67)	6.36 (19.69)
prompt: reasoning	implicit	16.67 [12.67, 20.67]	-5.56	47 : 22	4e-03	3.84 (4.52)	15.33 (17.56)	0.10 (0.11)	4.31 (6.18)	18.27 (25.87)	79.07 (80.80)	0.66 (0.67)	11.73 (19.69)
scrub: lexical*	implicit	0.67 [0.00, 1.56]	-21.56	98 : 1	3e-28	0.11 (4.52)	0.22 (17.56)	0.00 (0.11)	0.18 (6.18)	33.69 (25.87)	83.29 (80.80)	0.70 (0.67)	0.13 (19.69)
baseline (no mitigation)	explicit	24.44 [20.22, 28.67]				5.33	21.33	0.10	6.58	40.22	80.04	0.67	8.80
scrub: llm	explicit	0.67 [0.00, 1.56] ♦	-23.78 ♦	108 : 1	3e-31	0.11 (5.33) ♦	0.22 (21.33) ♦	0.00 (0.10) ♦	0.18 (6.58) ♦	35.02 (40.22)	83.20 (80.04)	0.70 (0.67)	0.00 (8.80) ♦
SFT (LoRA, merged)	explicit	6.89 [4.44, 9.56]	-17.56	92 : 13	8e-16	1.21 (5.33)	3.78 (21.33)	0.03 (0.10)	2.13 (6.58)	26.62 (40.22)	82.98 (80.04)	0.66 (0.67)	0.09 (8.80)
prompt: second_pass_verification	explicit	10.22 [6.67, 14.00]	-14.22	102 : 38	6e-08	2.17 (5.33)	6.00 (21.33)	0.20 (0.10)	2.67 (6.58)	2.84 (40.22)	68.53 (80.04)	0.60 (0.67)	19.24 (8.80)
prompt: structured_rubric	explicit	10.67 [7.55, 14.00]	-13.78	101 : 39	2e-07	2.45 (5.33)	9.56 (21.33)	0.07 (0.10)	2.93 (6.58)	14.84 (40.22)	78.84 (80.04)	0.65 (0.67)	7.78 (8.80)
prompt: fairness_constitution	explicit	10.89 [8.00, 13.78]	-13.56	80 : 19	4e-10	1.80 (5.33)	6.89 (21.33)	0.04 (0.10)	3.33 (6.58)	28.76 (40.22)	81.38 (80.04)	0.70 (0.67)	0.76 (8.80)
prompt: recruiter_guidelines	explicit	14.89 [11.56, 18.22]	-9.56	79 : 36	8e-05	3.40 (5.33)	12.89 (21.33)	0.08 (0.10)	4.18 (6.58)	23.87 (40.22)	82.18 (80.04)	0.69 (0.67)	4.93 (8.80)
LEACE (layer 16)	explicit	8.37 [5.20, 11.93] (own baseline 15.94, n=251)	-7.57	28 : 9	3e-03	4.48 (5.33)	15.52 (21.33)	0.13 (0.10)	3.72 (6.58)	12.49 (40.22)	83.95 (80.04)	0.70 (0.67)	3.82 (8.80)
prompt: ignore_personal_info	explicit	16.89 [13.33, 20.44]	-7.56	48 : 14	2e-05	2.99 (5.33)	11.11 (21.33)	0.06 (0.10)	4.62 (6.58)	42.04 (40.22)	80.44 (80.04)	0.69 (0.67)	1.56 (8.80)
prompt: zero_shot_cot	explicit	17.56 [13.78, 21.33]	-6.89	62 : 31	2e-03	4.05 (5.33)	15.33 (21.33)	0.10 (0.10)	4.80 (6.58)	27.96 (40.22)	83.60 (80.04) ♦	0.69 (0.67)	8.22 (8.80)
prompt: counterfactual_invariance	explicit	18.49 [14.48, 22.54]	-6.01	78 : 51	0.02	4.13 (5.33)	17.56 (21.33)	0.11 (0.10)	4.67 (6.58)	19.16 (40.22)	78.52 (80.04)	0.67 (0.67)	4.67 (8.80)
prompt: reasoning	explicit	20.22 [16.22, 24.22]	-4.22	42 : 23	0.02	4.66 (5.33)	18.89 (21.33)	0.10 (0.10)	5.69 (6.58)	32.27 (40.22)	82.13 (80.04)	0.67 (0.67)	3.47 (8.80)
scrub: lexical*	explicit	0.22 [0.00, 0.67]	-24.22	109 : 0	3e-33	0.07 (5.33)	0.22 (21.33)	0.00 (0.10)	0.04 (6.58)	33.96 (40.22)	83.38 (80.04)	0.70 (0.67)	0.09 (8.80)

Values: mitigated run, baseline in brackets, both on exactly the rows the run evaluated. Unstable sets: share with its own 95% CI (bootstrap over candidates); Δ pp: change against the baseline; bold coloured Δ = p < 0.05 (exact sign test on fixed vs broken sets, uncorrected). A run with parse failures is compared on its complete sets only; its own baseline on those sets is shown under its value. Italic ♦ = best value in the column within its condition (real mitigations only; hire rate and FS have no better direction).

Fixed : broken — each counterfactual set (one candidate-job pair, one condition, all attribute variants) paired with the same set at baseline. Fixed: unstable at baseline (the attribute alone changed the decision), stable after the mitigation. Broken: stable at baseline, unstable after — the mitigation introduced a new dependence on the attribute. Δ unstable sets = (broken – fixed) / sets; e.g. 58 : 57 means the method fixes about as many sets as it breaks.

* Lexical scrubbing is an oracle upper bound, not a comparable mitigation: it removes the attribute with the study's own injection templates, so every variant becomes character-for-character the attribute-free CV (100% of rows) and the model sees one identical prompt — zero instability by construction. Real CVs do not use our templates; read this row only as the ceiling. LLM scrubbing, which must find the attribute itself, is the realistic version.